



NATIONAL UNIVERSITY OF MODERN LANGUAGES

SOFTWARE QUALITY ENGINEERING Open Ended Lab

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COURSE: SOFTWARE QUALITY ENGINEERING

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Question 1 – Error, Fault, and Failure

Error

The developer forgot to include the uppercase letter check while writing the code. This is a human mistake made before any code was written.

Fault

The function `validatePassword()` returns true without checking for an uppercase letter. The missing check is:

```
bool hasUpper = false;
for (int i = 0; i < password.length(); i++)
    if (isupper(password[i])) hasUpper = true;
if (!hasUpper) return false;
```

Failure

When "abc12345" is passed, the program prints "Password Accepted" even though the password has no uppercase letter. This incorrect output at runtime is the failure.

Summary

Term	What It Means	In This Code
Error	Developer's thinking mistake	Forgot the uppercase requirement
Fault	Defect in the code	Missing <code>isupper()</code> check in <code>validatePassword()</code>
Failure	Wrong output at runtime	"Password Accepted" printed for "abc12345"

Question 2 – SQA, SQC, and SQP Activities

Part (a) – Missing Activities in ShopEase

- **Requirements Review:** Requirements were never verified, so incorrect logic reached production.
- **Test Planning:** No test plan existed. Critical flows like payments were never formally tested.
- **Code Reviews:** No peer reviews were done, so bugs went unnoticed.
- **Performance Testing:** System was never load tested, causing slow performance after launch.
- **Defect Tracking:** No bug tracking system was in place to manage reported issues.

Part (b) – 4 SQA Activities Before Release

1. **Requirements Review:** Review all requirements with stakeholders before coding begins to catch missing or unclear logic early.

- 2. Unit and Integration Testing:** Test each module individually and together, especially the payment and order modules.
- 3. Performance and Load Testing:** Stress-test the system with simulated traffic to find bottlenecks before go-live.
- 4. User Acceptance Testing (UAT):** Have real users test the full system end-to-end before release to catch functional gaps.

Question 3 – GQM Model for Payment Reliability

Goal

Improve the reliability of payment processing in ShopEase so that customers can complete transactions without failures.

GQM Table

Goal	Question	Metric
Improve payment reliability	How often are payments failing?	Payment Failure Rate = (Failed / Total) x 100
Improve payment reliability	How fast does a payment go through?	Average Transaction Response Time (seconds)
Improve payment reliability	How quickly are payment bugs fixed?	Mean Time to Resolve (MTTR) payment defects
Improve payment reliability	Is the payment system available?	Uptime Percentage per month
Improve payment reliability	Are bugs caught before reaching users?	Defect Escape Rate = Prod defects / Total defects

How Metrics Help in Decision-Making

If Failure Rate exceeds 2–3%, the payment logic needs immediate fixing. High Response Time signals a performance issue. High MTTR means bugs are being resolved too slowly. Low Uptime points to infrastructure problems. A high Defect Escape Rate means testing needs to improve. Together these metrics give the team clear, evidence-based actions.

Question 4 – Cost of Software Quality

Given Data

Category	Cost (\$)
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Prevention	\$20,000
Appraisal	\$30,000
Internal Failure	\$45,000
External Failure	\$80,000

Part (a) – Total CoSQ

Total CoSQ = \$20,000 + \$30,000 + \$45,000 + \$80,000

Total CoSQ = 175,000

Part (b) – Cost of Conformance

Cost of Conformance = Prevention + Appraisal = \$20,000 + \$30,000

Cost of Conformance = 50,000

Part (c) – Cost of Non-Conformance

Cost of Non-Conformance = Internal Failure + External Failure = \$45,000 + \$80,000

Cost of Non-Conformance = 125,000

Part (d) – Highest Cost

Category	Cost	% of Total	Type
Prevention	\$20,000	11.4%	Conformance
Appraisal	\$30,000	17.1%	Conformance
Internal Failure	\$45,000	25.7%	Non-Conformance
External Failure	\$80,000	45.7%	Non-Conformance
TOTAL	\$175,000	100%	—

External Failure is highest at \$80,000 (45.7%). This means defects were not caught before reaching customers, resulting in complaints, refunds, and reputation damage.

Part (e) – How Prevention Reduces Total Cost

Defects caught early are cheaper to fix. Investing in requirements reviews, code reviews, and test automation prevents bugs from reaching production. Spending \$30,000 more on prevention could cut external failure costs by \$40,000+, lowering total CoSQ overall.